

Chapter 1

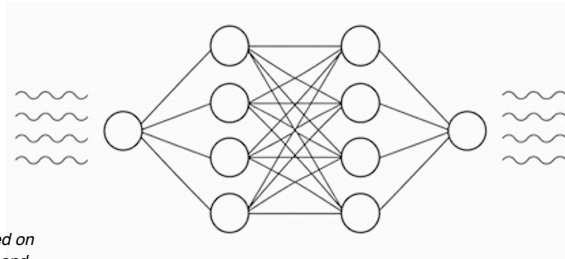
What is a Large Language Model?

Module 1: Foundations of Large Language Models

Welcome to our deep dive into Large Language Models. By the end of this chapter you'll have a mental model of what Large Language Models are, how they differ from what came before, and.. where they fall flat. Let's dig in.

LLM Definition

An artificial intelligence system trained on massive text datasets to understand and generate human language, typically containing billions of parameters and capable of performing diverse language tasks through pattern recognition and probabilistic text generation.



This is a diagram of a neural network — nodes connected by lines, with wavy lines representing data flowing in and out. That's essentially what an LLM is: a massive neural network trained on an enormous amount of text. Books, websites, forums, articles — basically the written output of humanity — fed through a network with billions or even trillions of parameters. Those parameters are the model's memory of everything it read.

LLMs versus NLP

How are they different?

NLPs

- Designed for specific tasks
- Trained specifically per task
- Thousands to millions of parameters

Use Cases

- Email spam filtering
- Language detection
- Document similarity scoring

Limitations

- Poor generalization
- Brittleness
- Limited creativity

LLMs

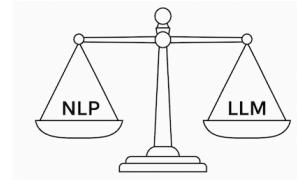
- Massive neural networks
- Vast data sets
- Natural language processing

Use Cases

- Content generation
- Customer support agents
- Code generation and debugging

Limitations

- Hallucinations
- Context length constraints
- Reasoning failures



I tried to come up with a better image but couldn't find one - so that scale on the right is doing a lot of work. NLP models and LLMs are both AI language tools — but they're not even playing the same sport.

NLP models were specialists — one for spam filtering, one for translation, each brittle and task-specific. None of them could create anything new. LLMs threw that playbook out. Train on one objective — predict the next word — at massive scale, and you get a generalist. But they have their own issues: hallucinations, context window limits, and reasoning failures. Powerful, but not infallible.

The Great Parameter Arms Race

How models got ridiculously large

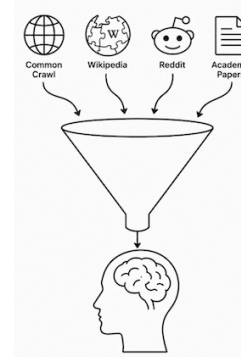
Model	Creator	Context Window	Intelligence Index	Price (USD/1M Tokens)	Output Tokens/s
Claude Opus 4.8	Anthropic	1M	61	\$4.10	58
GPT-5.5 (xhigh)	OpenAI	922k	60	\$4.35	62
GPT-5.5 (high)	OpenAI	922k	59	\$4.35	51
Claude Opus 4.7	Anthropic	1M	57	\$4.10	48
Gemini 3.1 Pro Preview	Google	1M	57	\$1.74	139
GPT-5.5 (medium)	OpenAI	922k	57	\$4.35	54
Qwen3.7 Max	Alibaba	1M	57	\$1.43	103
Gemini 3.5 Flash	Google	1M	55	\$1.31	177
MiniMax-M3	MiniMax	1M	55	\$0.22	40

I love that this slide is called "The Great Parameter Arms Race" because that's genuinely what happened.

Look at that progression. Millions of parameters, then 117 million, then 175 billion, then a rumored 1.7 trillion — all in about five years. That's not linear, that's exponential. And that scaling is what unlocked everything we'll talk about in this course.

Training Data Sources

- Trained on enormous datasets from the internet
- Involves hundreds of billions or even trillions of words
- Massive exposure to human language enables LLMs to understand context



That funnel image tells the story perfectly. All of this data — Common Crawl, Wikipedia, Reddit, academic papers — gets poured in at the top and the model absorbs it.

We're talking massive datasets, containing hundreds of billions to trillions of words. The Common Crawl dataset alone is petabytes of web content. That breadth of exposure is what gives LLMs their uncanny ability to understand context, nuance, and yes — even your niche pop culture references. They've read more than any human ever could.

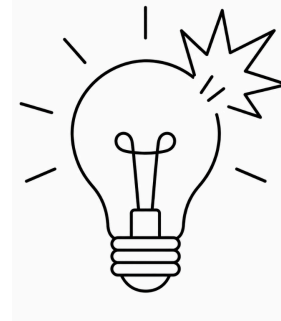


Someone's going to have to pay. Training GPT-3 cost around \$4.6 million in compute — just for one run. GPT-4 likely ran into the tens of millions. This is why only a handful of organizations can train frontier models from scratch. For the rest of us, the game is building on top of them — which is exactly what this course is about.

Emergent Capabilities

Surprise! Your AI Can Do That Now.

- Not explicitly programmed
- Organically grown
- Few-shot prompts
- Chain-of-thought reasoning
- Creative content



At a certain scale, models start doing things nobody programmed them to do. Few-shot learning, chain-of-thought reasoning, creative writing — these emerged organically from scale alone, not explicit design. We'll get very hands-on with these in the prompt engineering module.

LLM Families

Model	Creator	Context Window	Intelligence Index	Price (USD/1M Tokens)	Output Tokens/s
Claude Opus 4.8	Anthropic	1M	61	\$4.10	58
GPT-5.5 (xhigh)	OpenAI	922k	60	\$4.35	62
GPT-5.5 (high)	OpenAI	922k	59	\$4.35	51
Claude Opus 4.7	Anthropic	1M	57	\$4.10	48
Gemini 3.1 Pro Preview	Google	1M	57	\$1.74	139
GPT-5.5 (medium)	OpenAI	922k	57	\$4.35	54
Qwen3.7 Max	Alibaba	1M	57	\$1.43	103
Gemini 3.5 Flash	Google	1M	55	\$1.31	177
MiniMax-M3	MiniMax	1M	55	\$0.22	40

Here's the landscape as of mid-2026. Anthropic and OpenAI are neck and neck at the top. But the more interesting story is the challengers — Gemini at \$1.31 per million tokens, Qwen from Alibaba matching frontier intelligence at \$1.43. The market is more competitive than the headlines suggest.

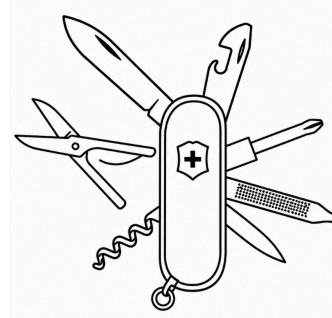
Capabilities



One model. Conversation, question answering, code generation, translation, summarization, creative content. All from that same simple objective: predict the next word.

Key Takeaways

- Paradigm shift from task-specific models
- General-purpose language systems
- Massive scale
- Emergent Capabilities



LLMs represent a genuine paradigm shift — from task-specific tools to general-purpose language systems. The secret ingredient is scale. And at sufficient scale, you get emergent capabilities nobody explicitly designed.